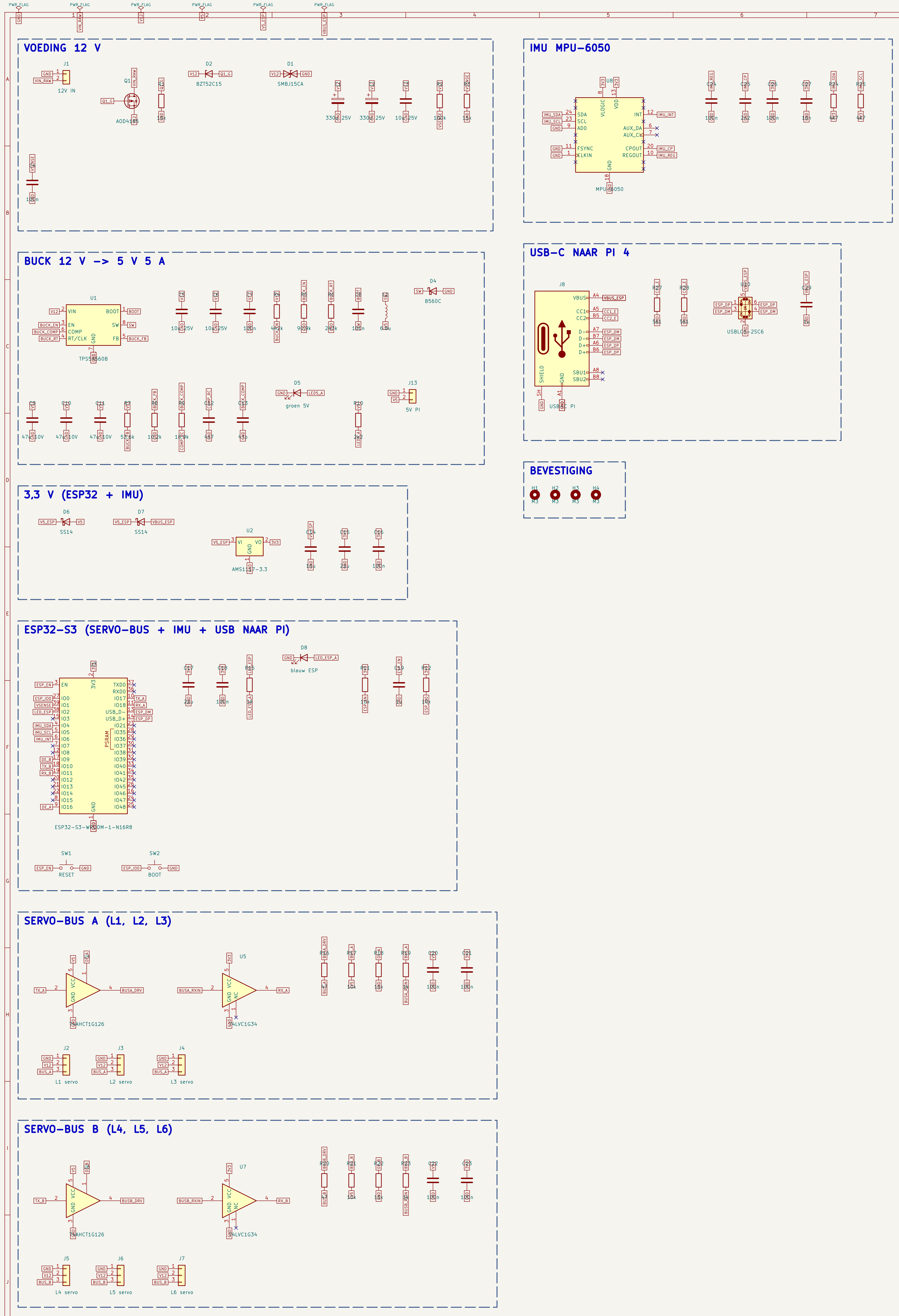

ARACHNE−18 MINI BRAIN v1.0 − ESP32−S3 + MPU−6050 hexapod servo controller for a Raspberry Pi 4 (generated by tools/build_sch.py from tools/design.py)
Servo connectors: 1 = GND, 2 = +12 V, 3 = DATA (Feetech STS half−duplex bus, 1 Mbaud). CHECK the pin order against your cable before plugging in!
Bus A = legs L1 L2 L3 (ESP32 UART1: TX IO17, RX IO18, DE IO16). Bus B = legs L4 L5 L6 (UART2: TX IO10, RX IO11, DE IO9). DE = RTS in RS485 half−duplex mode.
Pi 4 link: USB−C (board) <−> USB−A (Pi) cable on the ESP32−S3 native USB −> /dev/ttyACM0 on the Pi. Pi VBUS only feeds the ESP32 (through D7), never the servos.
5V PI pads: buck output (5 V, 3 A for the Pi 4) −> Pi GPIO pin 2/4 (+5V) and 6 (GND), or a cut USB−C lead into the Pi power socket.
Buck = TI TPS54560B datasheet design (fig. 32): 400 kHz, 5.0 V 5 A. UVLO start 6.5 V.

ESP32−S3 + MPU−6050, 6 servo−bus legs, USB to Pi 4
Kobe Van Robaeys − maakleerplek
Sheet: /
File: arachne_mini.kicad_sch
Title: ARACHNE−18 mini brain board
Size: A1
Date: 2026−10−18
Rev: 1.0
KiCad E.D.A. 10.0.3
Id: 1/1